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Theme: science
 other themes | news | about RNW

News

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Holland
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netherlinks
science
development
health
human rights
media
lifestyle
social issues

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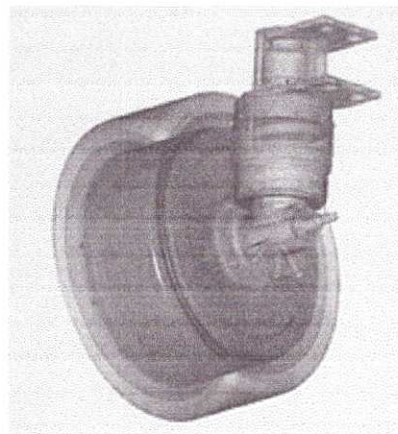
RN - the full story
partners
reactions
route
vacancies
press releases
products

The whispering wheel

by Thijs Westerbeek, 15 december 2003

A new Dutch invention can make cars, busses and other vehicles no less than 50 percent more efficient and thus more environmentally friendly. Better still, the technology is already available; it all comes down to a smart combination of existing systems.

This winter, in the city of Apeldoorn, a city bus will be used to prove that the claims about the new invention are true. These are quite bold. E-traction, the company that developed the bus, boasts fuel savings of up to 60 per cent, with emissions down to only a fraction of the soot and carbon dioxide an ordinary bus would blow out of its tailpipe.



This report was featured in Research File. Listen to the programme in full. (29:30)

In addition, the test bus requires no adaptation, its drivers need no extra training and there'll be no discomfort for passengers. It will simply run on diesel, just like all the other buses, and it should be just as reliable. One thing however will be very different; the Apeldoorn bus hardly makes a sound, hence its nickname "the whisperer".

In-wheel engine

All this is made possible by an 'in-wheel' electric engine, in fact nothing more than a normal electric engine turned inside out.

The outer wall of a traditional electric engine is a cylinder lined on the inside with copper wire. If electricity is fed into the copper wire, the current will circle the cylinder on the inside at high speed. Cylinder and wire together are called the 'stator' (because it doesn't move).

To change the electricity running along the inner wall of the cylinder into movement, another part of the engine comes into play: 'the rotor'. This is in fact an axle, mounted in the centre of the cylinder, with permanent magnets attached to it. The electrical current in the stator pulls the rotor magnets along and the axle starts to turn.



The wheel works precisely the other way around. The fixed part of the engine - the stator - is now on the inside. The wire is wrapped around it.

The moving part of the engine – the rotor – is no longer an axle fitted with magnets but a ring running on the outside of the stator.

The magnets are fixed on the inside of this ring. If power is fed into the engine the magnets will – as before – follow the current, but now it's the ring on the outside, which will turn.

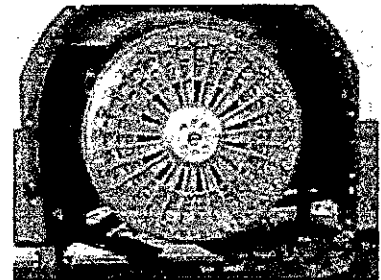
Eureka

And that's what makes 'the whisperer' revolutionary; a ring functioning as a wheel. By just putting a tire on it you can drive a bus, a car, anything with it. Since the wheel is in fact the engine, no axles or any other friction-producing and therefore energy-wasting mechanical parts are needed.

Even the transmission is unnecessary; if you want to go faster you just run more electricity through the engine. And it works really well while braking, when the in-wheel engine works as a generator, produces electricity to charge the batteries.

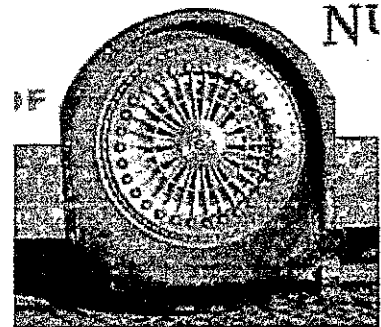
Pack of Batteries

The power to drive the Apeldoorn bus is stored in a big battery pack that sits in a steel drawer under the bus. Changing the batteries every time they're drained would be impractical, as would be taking the bus out of service for recharging them for hours on end. Instead, a small diesel-powered generator built into what used to be the bus's engine bay continuously charges the whole battery pack.



The whispering wheel with and without tyre

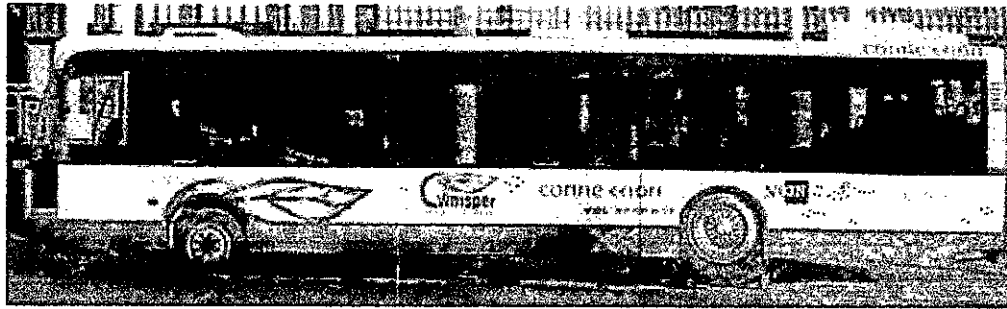
Since in-wheel engines are so highly efficient, the generator's diesel engine can be very small, about the size of the compact city car's engine. Because charging the batteries is all it needs to do, the tiny engine consumes very little fuel and can run continuously at a speed of 1700 revs per minute, the most efficient rev count.



Clean and quiet

Passengers will find it more important that the bus is quiet and clean. No more roaring buses pulling away from the station in a cloud of diesel fumes. When the whisperer pulls away (and whenever it drives for that matter), the power comes from the batteries, not the diesel engine which simply keeps on purring quietly.

Furthermore, the constant rev count makes the catalyser much more effective, and the small size of the engine makes it possible to completely fill the rest of the engine bay with sound proofing. Being 90 percent quieter than other buses, the 'whisperer' really deserves its name.



Testing period

In the coming six months the bus has to prove itself in everyday practice. Come summer, the city of Apeldoorn is set to decide whether to use whisperers on a larger scale in public transport. Dr Arjan Helnen, inventor of the whisperer and director of E-traction, radiates confidence: "This is a practical solution for present-day public transport. Every bus driver can get behind the wheel and do his job as before, only now it's quiet, clean and energy-efficient."

The future of the in-wheel electric engine seems bright. At the recent Tokyo Motor Show, it was the engine of choice in many of the futuristic hydrogen-powered concept cars.

Links

- Driving without moving (RNW)
- E-traction

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